可以把这张图解释成一句话先定调：

“This chart shows not just which features matter, but also which risk class each feature most strongly influences.”

然后分别解释三个轴/概念。

Overall Model Impact（X-axis）
这是这个 feature 对模型整体预测的影响强度。你这里用的是 **Sum of Mean |SHAP| Across Classes**，所以可以简单说：

“The further right a feature is, the more influence it has on the model’s overall decision.”

也就是说，越靠右，feature 越重要；它不代表 High 或 Low，只代表“模型有多在意它”。

Risk-Class Dominance（Y-axis）
这个是最需要讲清楚的。它不是“这个 feature 有多重要”，而是：

这个 feature 的 SHAP impact 有多集中在某一个 risk class 上。

比如一个 feature 的总 SHAP impact 是 1.0：

- High = 0.8
- Medium = 0.15
- Low = 0.05

那 High 占了 80%，dominance 就很高，说明这个 feature 主要是一个 **High-risk driver**。

但如果：

- High = 0.4
- Medium = 0.35
- Low = 0.25

虽然 feature 可能也很重要，但它没有明显“偏向”某一个 class，所以 dominance 会比较低。

你可以对领导说：

“The Y-axis tells us how class-specific the feature is. A high dominance means most of that feature’s model impact is concentrated in one risk class. A lower dominance means the feature influences multiple risk classes more evenly.”

所以你这张图本质上是：

- 右边 = **important**
- 上面 = **class-specific**
- 右上角 = **最值得关注的 feature**
    - 它既重要
    - 又明显推动某一个特定 risk class

这也是为什么这个图非常适合放在 **“Why the model assigned the risk rating”** 这一页。

Attention Boundary 怎么解释

这个虚线不要讲得太数学化。它的目的其实是帮助领导快速看哪些 feature 值得优先关注。

你可以说：

“The attention boundary is a visual prioritization threshold. Features above or to the upper-right side of the line combine relatively high model impact with strong class dominance, so these are the features we would investigate first when explaining a risk rating.”

也就是说，它不是模型算法本身的 decision boundary，也不是 classification threshold。

这个一定要避免领导误解。

建议明确说：

“This is not a model decision threshold. It is an interpretability threshold used to highlight the most meaningful explanatory features.”

你图里这个 boundary 本质是在 balancing 两个东西：

Impact × Dominance

一个 feature 即使 dominance 100%，但如果 impact 很低，也不一定重要。

反过来，一个 feature impact 很高，但 dominance 很低，说明它对多个 class 都有影响，也不一定能很好解释“为什么最终是 High”。

所以这个 boundary 就是在找：

high impact + high class specificity

你在 **presentation** 时可以直接这样讲

“On the X-axis, we measure overall model impact using SHAP. The further right a feature is, the more influence it has on the model’s prediction overall.

On the Y-axis, we measure risk-class dominance. This tells us how concentrated that feature’s impact is in one particular risk class. So a feature near 100% dominance is primarily associated with one class, while a feature lower on the chart affects several classes more evenly.

The features in the upper-right area are therefore the most useful for explaining the prediction, because they are both highly influential and strongly associated with a specific risk class.

The dotted attention boundary is simply an interpretability aid. It is not a model threshold; it helps us prioritize the features that have the strongest combination of overall impact and class-specific influence.”

然后指一下颜色：

“The bubble color tells us which risk class the feature primarily drives — red for High, orange for Medium, and green for Low.”

bubble size：

“And the bubble size tells us how many customers are affected by that feature.”

最后收口可以用一句非常 leadership-friendly 的：

“So this chart answers three questions at once: how important is the feature, which risk class does it primarily influence, and how broadly does it affect our customer population.”

这个就是你这张图最核心的价值。